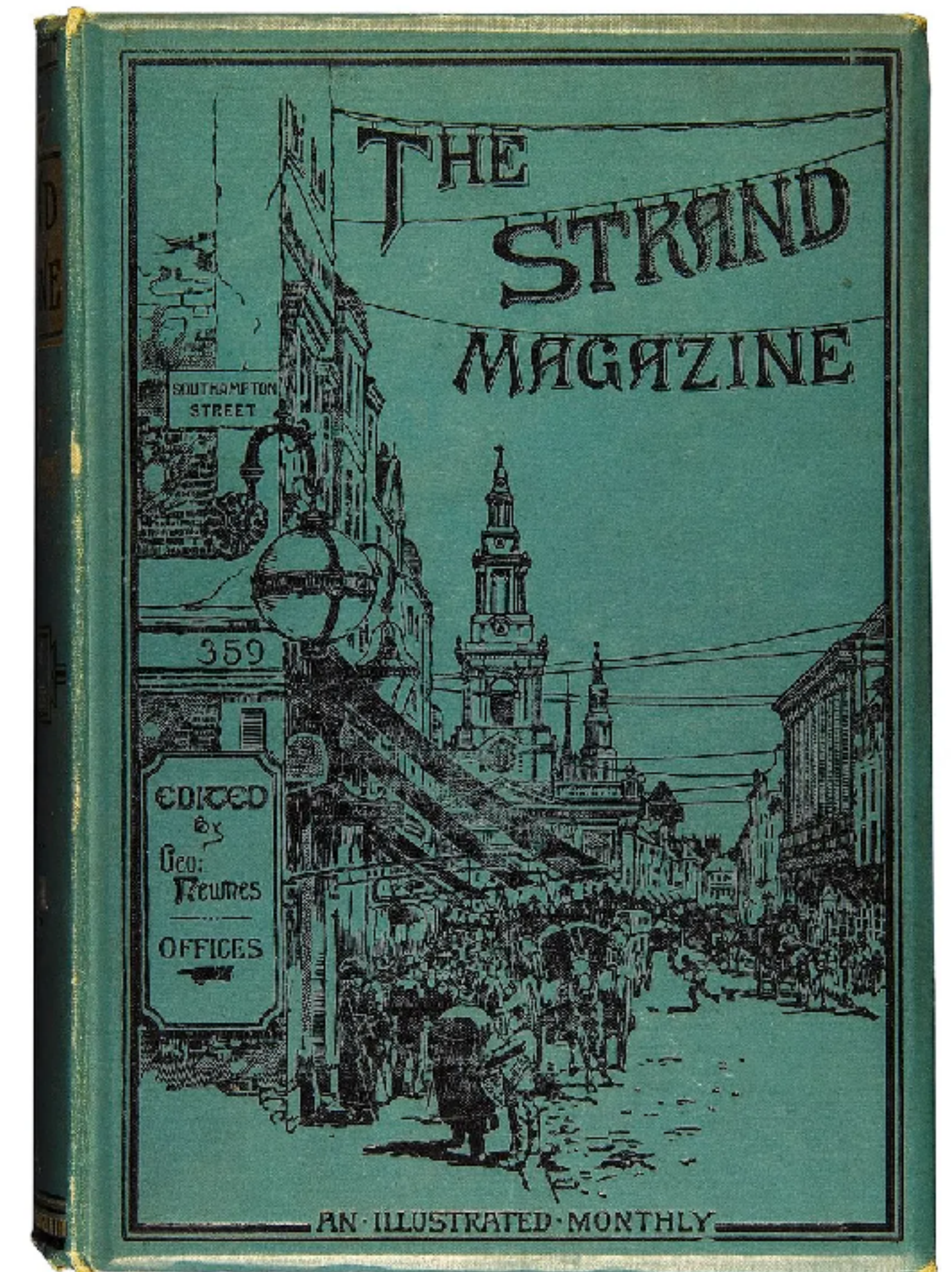


“I was talking the other day”, said William Rogers to the other villagers gathered around the inn fire, “to a gentleman about the place called Louvain. He said he knowed it well – used to visit a Belgian friend there. He said the house of his friend was in a long street, numbered on this side one, two, three and so on and that all the numbers of one side of him added up exactly the same as all the numbers on the other side of him. He said he knew there was more than 50 houses on that side of the street, but not so many as 500. Perhaps the reader may like to discover the number of that house.





1



2



3



4



5



6



7



8



1



2



3



4



5



6

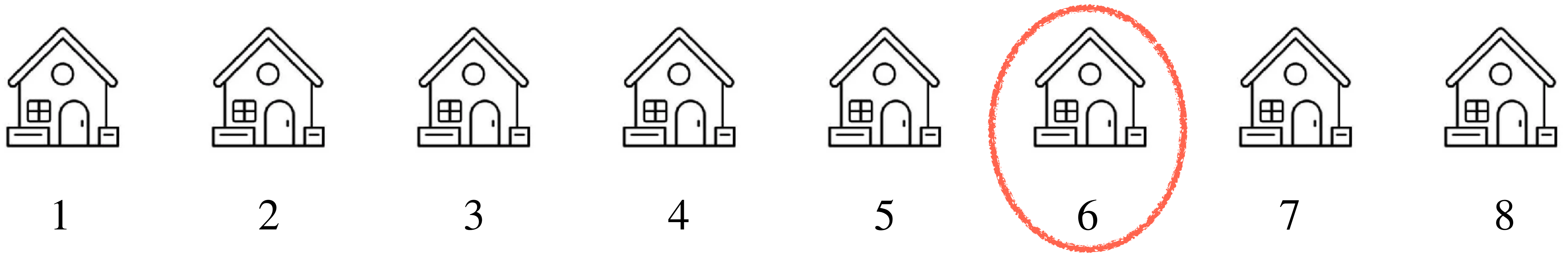


7



8

$$1 + 2 + 3 + 4 + 5 = 7 + 8$$



But the number of houses must be between 50 and 500.

$50 \leq m \leq 500$



1



2



3



4



5



6



7



8

n

m

$50 \leq m \leq 500$



1



2



3



...



$(n - 1)$



n

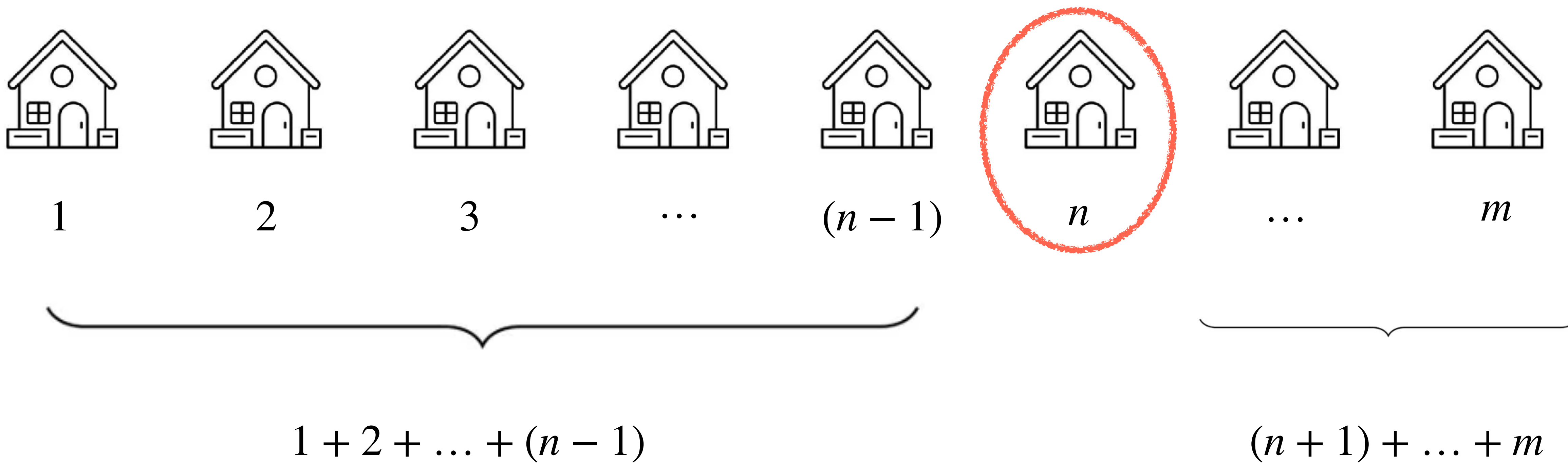


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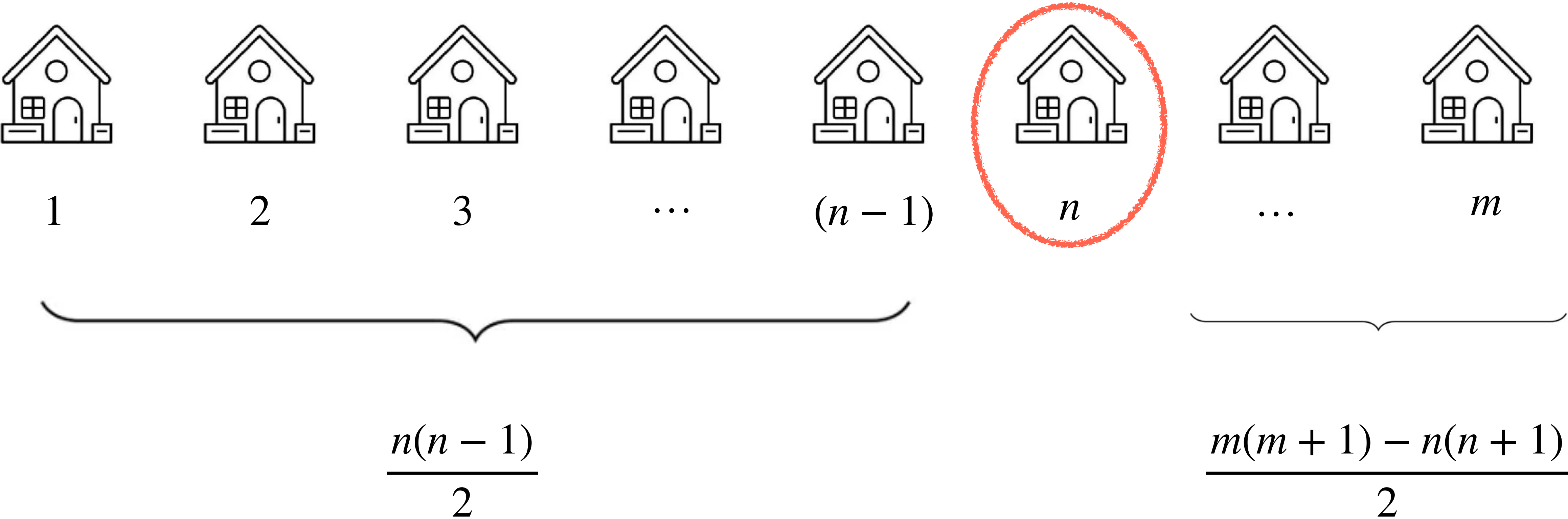


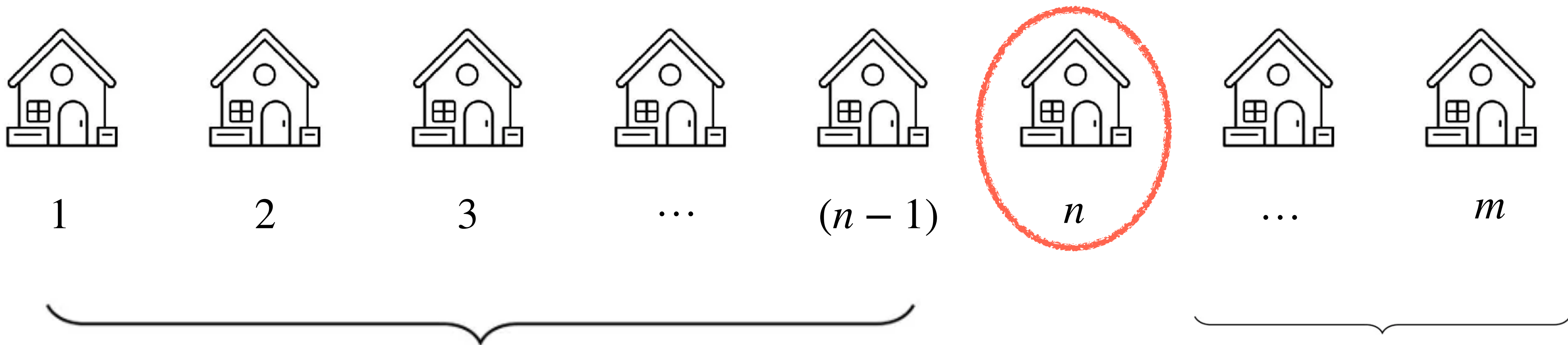
m

$$50 \leq m \leq 500$$



$$50 \leq m \leq 500$$





$$\frac{n(n-1)}{2} = \frac{m(m+1) - n(n+1)}{2}$$

$$50 \leq m \leq 500$$

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$$\frac{n(n-1)}{2} = \frac{m(m+1) - n(n+1)}{2}$$

$$50 \leq m \leq 500$$

$$n(n-1) = m(m+1) - n(n+1)$$

$$50 \leq m \leq 500$$

$$n(n - 1) = m(m + 1) - n(n + 1)$$

$$\implies n(n - 1) + n(n + 1) = m(m + 1)$$

$$50 \leq m \leq 500$$

$$n(n - 1) = m(m + 1) - n(n + 1)$$

$$\implies n(n - 1) + n(n + 1) = m(m + 1)$$

$$\implies 2n^2 = m^2 + m$$

$$50 \leq m \leq 500$$

$$n(n - 1) = m(m + 1) - n(n + 1)$$

$$\implies n(n - 1) + n(n + 1) = m(m + 1)$$

$$\implies 2n^2 = m^2 + m$$

$$\implies 8n^2 + 1 = 4m^2 + 4m + 1$$

$$50 \leq m \leq 500$$

$$n(n-1) = m(m+1) - n(n+1)$$

$$\implies n(n-1) + n(n+1) = m(m+1)$$

$$\implies 2n^2 = m^2 + m$$

$$\implies 8n^2 + 1 = 4m^2 + 4m + 1$$

$$\implies 8n^2 + 1 = (2m+1)^2$$

$$50 \leq m \leq 500$$

$$n(n-1) = m(m+1) - n(n+1)$$

$$\implies n(n-1) + n(n+1) = m(m+1)$$

$$\implies 2n^2 = m^2 + m$$

$$\implies 8n^2 + 1 = 4m^2 + 4m + 1$$

$$\implies 8n^2 + 1 = (2m+1)^2$$

$$\implies 8n^2 + 1 = k^2 \quad \text{where } k = 2m+1$$

$$50 \leq m \leq 500$$

$$n(n-1) = m(m+1) - n(n+1)$$

$$\implies n(n-1) + n(n+1) = m(m+1)$$

$$\implies 2n^2 = m^2 + m$$

$$\implies 8n^2 + 1 = 4m^2 + 4m + 1$$

$$\implies 8n^2 + 1 = (2m+1)^2$$

$$\implies 8n^2 + 1 = k^2$$

$$\text{where } k = 2m + 1$$

$$101 \leq k \leq 1001$$

When is $8n^2 + 1$ a perfect square?

Vargaprakṛti

When is $8n^2 + 1$ a perfect square?

Clearly, when $n = 0$

When is $8n^2 + 1$ a perfect square?

Clearly, when $n = 0$

Clearly, when $n = 1$

Given two solutions to the equation, how do I get more solutions?

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$$Dx^2 + z = y^2$$

$$Dx_1^2 + z_1 = y_1^2$$

$$Dx_2^2 + z_2 = y_2^2$$

Given two solutions to the equation, how do I get more solutions?

$$y^2 - Dx^2 = z$$

$$y_1^2 - Dx_1^2 = z_1$$

$$y_2^2 - Dx_2^2 = z_2$$

Given two solutions to the equation, how do I get more solutions?

$$y^2 - Dx^2 = z$$

$$(Dx_1^2 - y_1^2) (Dx_2^2 - y_2^2) = z_1 z_2$$

Given two solutions to the equation, how do I get more solutions?

$$y^2 - Dx^2 = z$$

$$D^2x_1^2x_2^2 - Dx_1^2y_2^2 - Dx_2^2y_1^2 + y_1^2y_2^2 = z_1z_2$$

Given two solutions to the equation, how do I get more solutions?

$$y^2 - Dx^2 = z$$

$$\begin{aligned} &+2Dx_1x_2y_1y_2 \\ D^2x_1^2x_2^2 - Dx_1^2y_2^2 - Dx_2^2y_1^2 + y_1^2y_2^2 &= z_1z_2 \\ &-2Dx_1x_2y_1y_2 \end{aligned}$$

Given two solutions to the equation, how do I get more solutions?

$$y^2 - Dx^2 = z$$

$$(y_1y_2 + Dx_1x_2)^2 - D(x_1y_2 + x_2y_1)^2 = z_1z_2$$

Given two solutions to the equation, how do I get more solutions?

$$Dx^2 + z = y^2$$

$$Dx_3^2 + z_3 = y_3^2$$

$$x_3 = x_1y_2 + x_2y_1, \quad y_3 = y_1y_2 + Dx_1x_2, \quad z_3 = z_1z_2,$$

Bhāvanā

If (x_1, y_1, z_1) and (x_2, y_2, z_2) are solutions to $Dx^2 + z = y^2$, then so is:

$$(x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$



Brahmagupta (c. 598 – c. 668 CE)

Bhāvanā

If (x_1, y_1, z_1) and (x_2, y_2, z_2) are solutions to $Dx^2 + z = y^2$, then so is:

$$(x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$

$$(x_1, y_1, z_1) \circ (x_2, y_2, z_2) = (x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$



Brahmagupta (c. 598 – c. 668 CE)

When is $8n^2 + 1$ a perfect square?

$(0, 1, 1)$ is a solution.

$(1, 3, 1)$ is a solution.

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$(0, 1, 1)$ is a solution.

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$$(x_1, y_1, z_1) \circ (x_2, y_2, z_2) = (x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$

When is $8n^2 + 1$ a perfect square?

$(0, 1, 1)$ is a solution.

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$$(x_1, y_1, z_1) \circ (x_2, y_2, z_2) = (x_1y_2 + x_2y_1, y_1y_2 + Dx_1x_2, z_1z_2)$$

$(1, 3, 1) \circ (1, 3, 1) = (6, 17, 1)$ is a solution.

When is $8n^2 + 1$ a perfect square?

$(0, 1, 1)$ is a solution.

$(1, 3, 1)$ is a solution.

$(6, 17, 1)$ is a solution.

When is $8n^2 + 1$ a perfect square?

$(0, 1, 1)$ is a solution.

$(1, 3, 1)$ is a solution.

$(6, 17, 1)$ is a solution.

$(6, 17, 1) \circ (6, 17, 1) = (204, 577, 1)$ is a solution.

$$50 \leq m \leq 500$$

$$\frac{n(n-1)}{2} = \frac{m(m+1) - n(n+1)}{2}$$

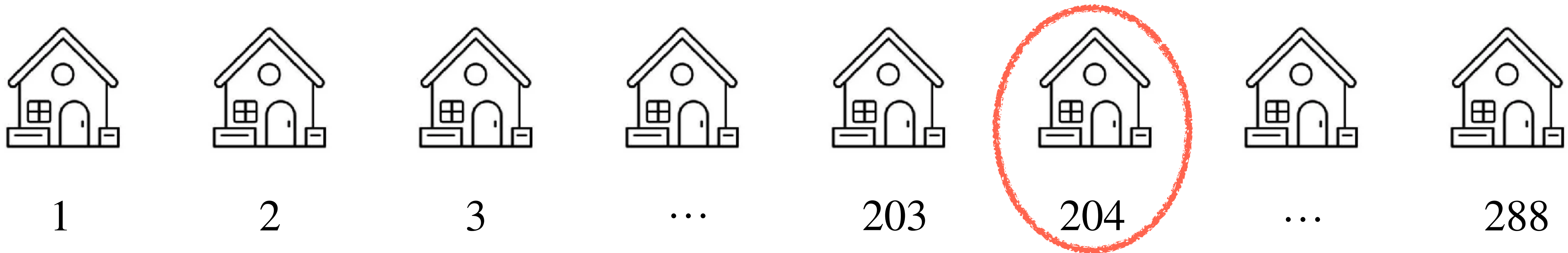
$$50 \leq m \leq 500$$

$$\frac{n(n-1)}{2} = \frac{m(m+1) - n(n+1)}{2}$$

$$n = 204$$

$$m = 288$$

$$50 \leq m \leq 500$$



$$1 + 2 + \dots + 203 = 205 + 206 + \dots + 288$$

कुर्वन्नावत्सराद् गणकः

$$92x^2 + 1 = y^2$$

One who can solve it within a year is truly a mathematician.

$$61x^2 + 1 = y^2$$

$$61x^2 + z = y^2$$

Always a solution: $(1, m, m^2 - 61)$

$$61x^2 + z = y^2$$

Always a solution: $(1, m, m^2 - 61)$

A possible solution: $(1, 8, 3)$

$$61x^2 + z = y^2$$

Always a solution: $(1, m, m^2 - 61)$

A possible solution: $(1, 8, 3)$

Composition: $(8 + m, 8m + 61, 3(m^2 - 61))$

$$61x^2 + z = y^2$$

If we divide by 3, we get:

$$\text{Composition: } \left(\frac{8 + m}{3}, \frac{8m + 61}{3}, \frac{3(m^2 - 61)}{9} \right)$$

$$61x^2 + z = y^2$$

If we divide by 3, we get:

$$\text{Composition: } \left(\frac{8 + m}{3}, \frac{8m + 61}{3}, \frac{m^2 - 61}{3} \right)$$

$$61x^2 + z = y^2$$

So let's take $m = 7$

Composition: $(5, 39, -4)$

$$61x^2 + z = y^2$$

Always a solution: $(1, m, m^2 - 61)$

A possible solution: $(5, 39, -4)$

Composition: $(39 + 5m, 39m + 305, -4(m^2 - 61))$

$$61x^2 + z = y^2$$

If we divide by -4 , we get:

$$\text{Composition: } \left(\frac{39 + 5m}{4}, \frac{39m + 305}{4}, \frac{m^2 - 61}{4} \right)$$

$$61x^2 + z = y^2$$

So let's take $m = 5$

Composition: $\left(\frac{39 + 5m}{4}, \frac{39m + 305}{4}, \frac{m^2 - 61}{4} \right)$

$$61x^2 + z = y^2$$

step	m	x	y	z
0		1	8	3
1	7	5	39	-4
2	9	21	164	-5
3	6	58	453	5
4	9	195	1523	4
5	7	722	5639	-3
6	8	3805	29718	-1

$$61x^2 + z = y^2$$

$$(3805, 29718, -1) \circ (3805, 29718, -1) = (226153980, 1766319049, 1)$$

Talk about the Fermat context.

How to select m ?

Prove that the product of two integers, each of which is the sum of two perfect squares, can itself always be expressed as the sum of two perfect squares.

dsShow that the product of any four consecutive integers
written as a difference of two perfect squares

Let $P(x)$ be a polynomial with real coefficients such that $P(x) \geq 0$ for all $x \in \mathbb{R}$. Prove that there exist polynomials $Q_1(x)$ and $Q_2(x)$ with real coefficients such that

[

$$P(x) = Q_1(x)^2 + Q_2(x)^2 \quad \text{for all } x \in \mathbb{R}.$$

]

Remark. This is a special case of Hilbert's seventeenth problem (solved by Artin in 1927~\cite{Artin1927}).